Physics 468/668: Statistical Physics **Spring 2019**

Lecture 11: The Grand Canonical Ensemble

Lecturer: Qiang Zhu *Scribes: scribe-name1,2,3*

11.1 Background

We established the basis of ensemble theory and made a detailed study of the micro-canonical and canonical ensembles. However, for a number of problems, the usefulness of them turn out to be rather limited and it appears that a further generalization of this formalism is called for. From micro-canonical to canonical ensemble, the macro-state of the systems was defined from a fixed (N, V, E) to (N, V, T). We can continue to expand them to a state where both N and E are variables.

11.2 Equilibrium between a system and a particle-energy reservoir

We consider the system A as immersed in a large reservoir A′, with which it can exchange both energy and particles. After some time has elapsed, the system and the reservoir are supposed to attain a state of mutual equilibrium. Then, they will reach a common temperature T and chemical potential μ. The fraction of the total number of particlesN^0 and the fraction of the total energy E^0 are conserved. While the system can take any energy E_r and number of particles N_r anywhere between zero and unity. Therefore, it satisfies,

$$N_r + N_r^{'} = N^0 \quad \text{const} \tag{11.1}$$

$$E_r + E_r^{'} = E^0 \quad \text{const} \tag{11.2}$$

Since the reservoir is supposed to be much larger than the system,

$$N_r \ll N_r^{'} \quad \text{const} \tag{11.3}$$

$$E_r \ll E_r^{'} \quad \text{const} \tag{11.4}$$

Now, in the manner of Lec 07, the probability $P_{r,s}$ that at any time t, the system A is found to be in an (N_r, E_s) state would be directly proportional to the number of micro-states $\Omega'(N_r', E_s')$ that the reservoir can have for the corresponding macro-state (N_r', E_s') . Thus,

$$P_{r,s} \propto \Omega'(N^0 - N_r, E^0 - E_s) \tag{11.5}$$

This can be written as

$$\begin{aligned} \ln \Omega'(N^0 - N_r, E^0 - E_s) &= \ln \Omega'(N^0, E^0) + \frac{\partial \ln \Omega'}{\partial N'}_{N'=N^0}(-N_r) + \frac{\partial \ln \Omega'}{\partial E'}_{E'=E^0}(-E_s) + ... \\ &\approx \ln \Omega'(N^0, E^0) + \frac{\mu'}{kT'} N_r + \frac{1}{kT'} E_s \end{aligned} \tag{11.6}$$

Therefore, we obtain the desired result:

$$P_{r,s} \propto \exp(-\alpha N_r - \beta E_s) \tag{11.7}$$

where

$$\alpha = \mu/kT, \quad \beta = 1/kT \tag{11.8}$$

By normalization, it becomes,

$$P_{r,s} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.9}$$

The summation in the denominator goes over all the (N_r, E_s) states accessible to the system A.

11.3 A system in the grand canonical ensemble

We now consider an ensemble of $\mathcal{N}$ identical systems mutually sharing a total number of particles $\mathcal{N}\bar{N}$ and a total energy of $\mathcal{N}\bar{E}$. Let $n_{r,s}$ denote the number of states of the systems that have, at any time t, the number of N_r of particles and amount of E_s of energy (r, s = 0, 1, 2, ...) then follows,

$$\begin{aligned} \sum_{r,s} n_{r,s} &= \mathcal{N} \\ \sum_{r,s} n_{r,s} N_r &= \mathcal{N}\bar{N} \\ \sum_{r,s} n_{r,s} &= \mathcal{N}\bar{E} \end{aligned} \tag{11.10}$$

Any set $\{n_{r,s}\}$ of the numbers $n_{r,s}$ which satisfies the restrictive conditions, represents one of the possible modes of distribution of particles and energy among the members of our ensemble. Furthermore, any such mode of distribution can be realized in different ways, the number is

$$W\{n_{r,s}\} = \frac{\mathcal{N}!}{\prod_{r,s}(n_{r,s}!)} \tag{11.11}$$

We may now define the most probable mode of distribution $n^*_{r,s}$ as the one that have maximum W. Following the way which we did on canonical ensemble, we obtain for a large ensemble,

$$\frac{n^*_{r,s}}{\mathcal{N}} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.12}$$

Alternatively, we may define the expectation values of the numbers $n_{r,s}$ as

$$< n_{r,s} >= \frac{\sum_{\{n_{r,s}\}} n_{r,s} W\{n_{r,s}\}}{\sum_{\{n_{r,s}\}} W\{n_{r,s}\}} \tag{11.13}$$

where the primed summations go over all distributions sets that conforms to the conditions. When the number $\mathcal{N}$ is very large, we find

$$\lim_{N\to\infty} \frac{< n_{r,s} >}{\mathcal{N}} \approx \frac{n^*_{r,s}}{\mathcal{N}} = \frac{\exp(-\alpha N_r - \beta E_s)}{\sum_{r,s} \exp(-\alpha N_r - \beta E_s)} \tag{11.14}$$

in agreement with the equation with we derived from the macroscopic picture.

11.4 Physical quantities

To establish the connection between the statistics of the grand canonical ensemble and the thermodynamics of the system, we introduce a quantity q,

$$q = \ln\big[\sum_{r,s} \exp(-\alpha N_r - \beta E_s)\big] \tag{11.15}$$

the quantity q is a function of α, β and also all the E_s. Taking the differential of q and making use of equations, we get

$$dq = -\bar{N}d\alpha - \bar{E}d\beta - \frac{\beta}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s \tag{11.16}$$

so that

$$d(q + \alpha\bar{N} + \beta\bar{E}) = \beta\big(\frac{\alpha}{\beta} + d\bar{E} - \frac{1}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s\big) \tag{11.17}$$

To interpret the terms, we recall the first law of thermodynamics,

$$\delta Q = -\mu d\bar{N} + d\bar{E} + \delta W \tag{11.18}$$

where the various symbols have their usual meanings.

$$\delta W = -\frac{1}{\mathcal{N}}\sum_{r,s} < n_{r,s} > dE_s, \;\; \mu = -\frac{\alpha}{\beta} \tag{11.19}$$

with the result that

$$d(q + \alpha\bar{N} + \beta\bar{E}) = \beta\delta Q \tag{11.20}$$

the quantity $(q + \alpha\bar{N} + \beta\bar{E})$ would be then identified with the thermodynamic variable S/k ,

$$q = \frac{S}{k} - \alpha\bar{N} - \beta\bar{E} = \frac{TS + \mu\bar{N} - \bar{E}}{kT} = \frac{G - (E - TS)}{kT} = \frac{PV}{kT} \tag{11.21}$$

To derive further results, we also defined

$$z = e^{-\alpha} = e^{\mu/kT} \tag{11.22}$$

the parameter z is referred to fugacity of the system, In terms of z, the $q-$potential takes the form

$$q = \ln\big[\sum_{r,s} z^{N_r} e^{-\beta E_s}\big] = \ln\big[\sum_{r,s} z^{N_r} Q_{N_r}(V,T)\big] \tag{11.23}$$

So we can write

$$q(z,V,T) = \ln \mathcal{Q}(z,V,T) \tag{11.24}$$

where $\mathcal{Q}$ is the grand partition function

$$\mathcal{Q}(z,V,T) = \sum_{N_r=0}^{\infty} Z^{N_r} Q_{N_r}(V,T) \tag{11.25}$$

With these quantities being defined, we can now derive the leading thermodynamic quantities,

$$P(z,V,T) = \frac{kT}{V}q(z,V,T) = \frac{kT}{V}\ln \mathcal{Q}(z,V,T) \tag{11.26}$$

$$N(z,V,T) = z\left[\frac{\partial}{\partial z}q(z,V,T)\right]_{V,T} = kT\left[\frac{\partial}{\partial \mu}q(z,V,T)\right]_{V,T} \tag{11.27}$$

$$U(z,V,T) = -\left[\frac{\partial}{\partial \beta}q(z,V,T)\right]_{V,T} = kT^2\left[\frac{\partial}{\partial T}q(z,V,T)\right]_{V,T} \tag{11.28}$$

The Helmholtz free energy can be expressed as,

$$A = N\mu - PV = NkT\ln z - kT\ln \mathcal{Q}(z,V,T) = -kT\ln\frac{\mathcal{Q}(z,V,T)}{z^N} \tag{11.29}$$

which may be compared with the canonical ensemble

$$A = -kT\ln Q(N,V,T) \tag{11.30}$$

Finally, the entropy is

$$S = \frac{U-A}{T} = kT\left(\frac{\partial q}{\partial T}\right)_{z,V} - Nk\ln z + kq \tag{11.31}$$